

Nmap: Network Scanning for Open Ports

Assignment study PDF — English

1. What is Nmap?

Nmap is an open-source network discovery and security auditing tool. It can identify live hosts, open/closed/filtered ports, services, versions and other network information.

2. Basic scan

`nmap 192.168.1.10` scans a target host using Nmap's default behavior.

3. Specific ports

`nmap -p 22,80,443 192.168.1.10` scans selected ports. `nmap -p 1-1000 192.168.1.10` scans a port range. `nmap -p- 192.168.1.10` requests a scan of all TCP ports.

4. Service detection

`nmap -sV 192.168.1.10` attempts to determine the service and version associated with discovered ports.

5. Scan types

A TCP connect scan can be requested with `-sT`. A SYN scan can be requested with `-sS` and normally requires elevated privileges. UDP scanning uses `-sU`.

6. Reading results

Open means an application is listening. Closed means the host is reachable but no application is listening on that port. Filtered means a firewall or packet filter prevents Nmap from determining the port state.

7. Ethics

Only scan systems and networks you own or have explicit permission to test. Use Nmap in your own VMs or a permitted lab.